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PROPOSAL

**DESIGN AND DEVELOPMENT OF A USER-ECCENTRIC SOCIAL
MEDIA ANALYTIC TOOL FOR SMEs**

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CANDIDATE'S DECLARATION

We, the undersigned, hereby declare that this Dissertation is the result of our own original research work carried out independently and through our joint effort. We affirm that no part of this work has been submitted, either in whole or in part, for the award of any other degree or academic qualification in this or any other University.

We further declare that all materials, ideas, data, and information derived from the work of other scholars, authors, researchers, and sources have been fully acknowledged in accordance with academic and ethical standards. Every citation, reference, and borrowed concept used in the preparation of this Dissertation has been clearly indicated.

We take full responsibility for the authenticity and originality of the content presented, and we confirm that the research findings, analyses, and conclusions reflect our own intellectual effort and understanding.

Signed: _____

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SUPERVISOR'S DECLARATION.

I certify that the preparation and the presentation of the Dissertation were in accordance with the guidelines on supervision of Dissertations laid down by the University of Professional Studies, Accra.

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DATE:

DEDICATION

This project is first and foremost dedicated to Almighty God, whose grace, wisdom, and strength guided us through every stage of this work. Without His protection, favour, and guidance, this accomplishment would not have been possible. To Him be all the glory.

We also dedicate this project to our lecturers and academic mentors, whose knowledge, encouragement, and unwavering commitment to our growth have inspired us throughout this journey. Their combined influence has shaped our understanding, strengthened our perseverance, and made this achievement truly possible.

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INTRODUCTION

Digital technologies have revolutionized how people communicate and engage with organizations, with social media becoming a key platform for interaction, marketing, and public communication. The massive flow of posts, comments, and reactions provides organizations with real-time behavioural data, offering insights into customer interests, brand perception, and emerging trends.

To stay competitive, organizations increasingly rely on social media analytics tools like Google Analytics, Hootsuite, Brandwatch, Sprout Social, and Meta Business Suite for real-time monitoring, sentiment analysis, audience profiling, and campaign evaluation. However, many, especially in developing economies, face challenges such as limited analytical skills, high software costs, and reliance on basic metrics like likes and follower counts. This study examines these tools, their uses, benefits, and challenges, demonstrating how effective social media analytics can enhance marketing performance, organizational learning, and competitive advantage.

BACKGROUND OF THE STUDY

Social Media Analytics in Organizational Decision-Making

Social media has emerged as one of the most influential digital ecosystems, generating vast amounts of user-generated data every second through posts, comments, likes, shares, and multimedia content. Platforms such as Facebook, Instagram, TikTok, YouTube, LinkedIn, and X (formerly Twitter) play a major role in communication, customer engagement, brand visibility, and public opinion formation. This data-driven environment presents organizations with opportunities to understand audience behaviour, monitor performance, and make informed decisions in real time. As noted by Stieglitz et al. (2018), social media has become a critical source of organizational intelligence due to the richness and immediacy of its data. Despite this, many organizations particularly in developing economies struggle to interpret and utilize social media data effectively. Studies by Boadi, He, and Bosompem (2022) reveal that businesses often rely on basic metrics or intuition rather than structured analytical tools, leading to poor marketing evaluation and limited understanding of consumer preferences. This gap prevents organizations from fully leveraging opportunities to improve engagement, anticipate trends, and enhance strategic decision-making.

To address this challenge, social media analytics tools such as Google Analytics,

Hootsuite, Brandwatch, Sprout Social, and Meta Business Suite have become essential for tracking engagement, analyzing sentiment, monitoring trends, and evaluating campaigns. These tools leverage Big Data Analytics principles, focusing on the volume, velocity, variety, veracity, and value of digital data (Rocco et al., 2020). Social Network Theory further explains how users form connections, influence each other, and shape information flows that analytics tools measure using metrics like reach, impressions, and engagement (He, Zha & Li, 2013).

Key concepts underpinning this study include social media as platforms enabling interactive content sharing (Chae, 2015), social media data analytics as the systematic collection and analysis of content for decision-making (Stieglitz et al., 2018), and analytics tools as specialized systems providing dashboards and metrics to interpret user behaviour (Wang & Zhang, 2023). Sentiment analysis, competitive intelligence, and monitoring dashboards collectively support the technological foundation upon which organizations build digital strategies.

The growing reliance on social media for communication and marketing, combined with the limited adoption of analytics tools, highlights the importance of research exploring their functions, benefits, and challenges. Understanding these tools enables organizations to improve decision-making, strengthen competitive

intelligence, and maximize the value of the vast digital data generated daily across social platforms.

PROBLEM STATEMENT:

Challenges in Social Media Data Analysis

Many organizations actively engage on social media, yet a significant number struggle to understand and analyze the enormous volume of data generated on these platforms. Businesses often rely on basic metrics such as likes, comments, or follower counts to assess their online performance, but these indicators provide limited insight into actual customer sentiment or the effectiveness of marketing strategies. According to Boadi and Bosompem (2022), numerous firms in Ghana still depend on guesswork to evaluate their social media presence due to a lack of tools and expertise necessary to correctly interpret digital data.

This situation presents a considerable challenge because social media continuously produces streams of information, including customer opinions, behavioural patterns, trending topics, and engagement signals, which could be invaluable for informed decision-making. Without proper analytics tools, however, much of this data is overlooked or misinterpreted. As noted by Stieglitz et al. (2018), social media data only becomes valuable when systematically collected and analyzed. Without structured analysis, organizations risk making poor marketing decisions, wasting resources on ineffective campaigns, and failing to respond to shifts in customer interests and preferences.

The challenge is further compounded by factors such as limited digital literacy, insufficient training, and the high cost of advanced analytics platforms, particularly for small and medium-sized enterprises in developing economies (Wang & Zhang, 2023). As a result, organizations invest considerable time and money in social media activities but fail to extract strategic value or actionable insights from the generated data.

Given these constraints, there is a clear need to explore and understand the range of tools available for effective social media data analysis. Investigating these tools can help demonstrate how they enable organizations to systematically collect, monitor, and interpret digital interactions. Effective use of analytics tools can support more accurate decision-making, improve the understanding of customer engagement, identify emerging trends, and optimize marketing performance. By addressing these gaps, organizations can transform raw social media interactions into actionable intelligence that enhances strategic planning, strengthens competitive positioning, and maximizes the return on investment from social media initiatives.

Moreover, the study of social media analytics tools highlights the broader importance of building digital capabilities within organizations. Training staff to use analytics platforms effectively, integrating data-driven processes into organizational workflows, and adopting cost-effective solutions can collectively

overcome current limitations. As businesses increasingly operate in data-rich environments, the ability to interpret social media insights becomes critical for sustaining competitive advantage, improving customer relationships, and making marketing campaigns more responsive and impactful.

In summary, the problem lies not in the lack of social media data but in the inability of organizations to leverage it meaningfully. By focusing on the adoption and effective use of analytics tools, this study seeks to address the knowledge gap, showing how organizations can turn digital information into strategic value, drive informed marketing decisions, and achieve enhanced business outcomes.

RESEARCH OBJECTIVES

GENERAL OBJECTIVE

The main objective of this study is to examine the social media data analytics tools available to organizations and evaluate how these tools support data-driven decision-making, improve customer engagement measurement, and enhance overall marketing performance (Stieglitz et al., 2018).

SPECIFIC OBJECTIVES

- To identify the major social media data analytics tools used locally and internationally.
- To examine the key features, capabilities, and technical functions of these tools.
- To analyze the benefits organizations gain from using social media analytics tools.
- To explore the challenges organizations face when adopting or using social media analytics tools.
- To provide recommendations on how organizations can strengthen their use of social media data analytics.

RESEARCH QUESTION

What are the major social media data analytics tools used by organizations both locally and internationally?

- What are the key features, capabilities, and technical functions of these social media analytics tools?
- What benefits do organizations gain from using social media analytics tools in their operations and marketing strategies?
- What challenges do organizations face when adopting or using social media analytics tools, particularly in developing economies?
- How can organizations improve and strengthen their use of social media data analytics to enhance decision-making and performance?

1.4.SIGNIFICANCE OF STUDY

Theoretically, this study contributes to understanding how Big Data Analytics and Social Network Theory are applied in analyzing social media data. Big Data Analytics Theory emphasizes the need to manage the volume, variety, and velocity of digital information (Rocco et al., 2020), while Social Network Theory explains how online interactions shape influence and communication flow among users (He, Zha & Li, 2013). By examining how different analytics tools incorporate these principles, the study provides insights into data-driven decision-making, audience behaviour, and digital engagement patterns. For instance, sentiment analysis supports Big Data concepts by processing large sets of user comments, network graphs in tools like Brandwatch illustrate Social Network Theory through influencer mapping, and real-time dashboards reflect the “velocity” component of Big Data Analytics Theory.

Academically, the study serves as a valuable reference for students, lecturers, and researchers interested in interpreting social media data using analytics tools. It addresses the need highlighted by Chae (2015) for more academic work explaining how unstructured online data can be transformed into meaningful insights. The

study clarifies processes such as data mining, trend detection, and engagement measurement, making it a useful resource for assignments, research papers, case studies, and future projects in information systems, marketing, and data science. Specifically, students researching digital marketing can learn how audience metrics are interpreted, researchers can cite the comparative analysis of tools like Hootsuite, Sprout Social, and Google Analytics, and lecturers can incorporate explanations of sentiment analysis and trend detection into teaching materials. Practically, the study benefits organizations and marketing teams that rely on social media for customer engagement and brand promotion. Many firms lack the skills or systems to effectively interpret social media data, resulting in weak marketing decisions (Boadi, He & Bosompem, 2022). By outlining the functions, advantages, and limitations of various analytics tools, the study aids businesses in selecting appropriate platforms and applying insights to improve marketing performance, understand customer needs, and make better strategic decisions. For example, businesses can use analytics dashboards to determine optimal posting times for engagement, marketing teams can leverage sentiment analysis to address customer complaints, managers can track competitor performance online to refine strategies, and SMEs can identify affordable tools such as Meta Business Suite to monitor engagement without incurring high subscription costs.

1.5.SCOPE OF THE PROJECT:

This project is limited to reviewing existing social media data analytics tools and designing a simplified software solution that demonstrates how organizations can analyse basic social media engagement data. The scope includes two main components: a literature review and the design and development of a prototype analytics system. The literature review will cover social media platforms, data analytics concepts, existing analytics tools, and the challenges businesses face when interpreting online engagement data. It will rely entirely on secondary data sources such as journals, books, online articles, and tool documentation.

The software development component will involve designing and implementing a small-scale prototype that demonstrates selected features of a social media analytics system. The system will not operate as a fully integrated commercial analytics platform; rather, it will serve as an academic prototype to show how social media data can be processed, visualized, and interpreted. The prototype will include key modules such as a user login interface, a data input/upload module, a basic data analysis module, and a dashboard for visualizing simple engagement

metrics. These metrics may include likes, comments, shares, sentiment indicators, and trend summaries based on input datasets.

The project will not include the development of a real-time data-harvesting engine, integration with external APIs such as Facebook Graph API or Twitter API, or advanced machine-learning models for sentiment analysis. It also will not cover large-scale database architectures, predictive modelling, enterprise-level security features, or full mobile app development. The system will only process sample datasets manually uploaded by the user for demonstration purposes, and will not collect live data from external social media platforms.

Additionally, the project will not evaluate every social media analytics tool currently available. Instead, it will focus on a selected number of tools for comparative analysis in the literature review. The system itself will remain a simplified academic model designed solely for the purpose of demonstrating core concepts discussed in the research. The supervisor and examiner will evaluate the project based on the stated modules, features, and the accuracy of the implemented prototype in meeting the defined scope.

1.6.LIMITATIONS OF THE STUDY

This study has several limitations that may influence the depth and outcomes of the research. First, it relies solely on secondary data sources such as published articles, books, and online reports, without primary data from interviews or surveys. This limits firsthand insights into how businesses in Ghana use social media analytics tools, as noted by Boadi, He, and Bosompem (2022).

Second, the study focuses on a selected number of popular analytics tools Google Analytics, Hootsuite, Sprout Social, Brandwatch, and Meta Business Suite excluding emerging or specialized platforms, which may reduce the variety of tools examined. The prototype developed is also a simplified academic system and not a fully functional commercial platform; it analyses manually uploaded sample datasets rather than integrating with real-time APIs. As Sommerville (2016) explains, academic prototypes often prioritise demonstrating core concepts over full functionality. Additionally, the prototype does not include advanced security features, predictive analytics, or large database infrastructures, which limits its real-world applicability. Constraints in development tools, computing resources, and time also restrict the complexity of features implemented. Despite these

limitations, the study and prototype still provide meaningful insights into how social media analytics tools operate and how organizations can benefit from them.

1.7.ORGANIZATION OF THE STUDY

This project report is divided into five chapters to guide the reader through the research and prototype development process. Chapter One introduces the study, outlining the background, problem statement, research questions, objectives, significance, scope, and limitations. Chapter Two reviews relevant literature, covering key theories, previous studies on social media analytics tools, and the conceptual framework. Chapter Three explains the methodology, highlighting the Waterfall Model used for system development, including requirements, design, implementation, testing, and documentation. Chapter Four details the design and development of the software prototype, describing system modules, interfaces, and testing results to show functionality and performance. Chapter Five concludes with a summary of findings, conclusions, recommendations, and reflections on limitations, emphasizing the study's contributions to academic knowledge and practical application.

CHAPTER TWO

2.1.THEORETICAL FRAMEWORK

This study is grounded in several theoretical models that help explain how social media data is generated, processed, interpreted, and adopted by organizations.

These theories provide the basis for understanding how analytics tools function and why organizations either embrace or struggle to use them effectively. The framework brings together concepts from information systems, data analytics, and digital communication to support the arguments and direction of the study.

The first guiding theory is Big Data Analytics Theory, which focuses on the management and analysis of extremely large and fast-moving datasets. Social media platforms generate massive volumes of real-time data that exhibit the “5Vs” of Big Data: volume, velocity, variety, veracity and value (Rocco et al., 2020). Big Data Analytics Theory explains how specialized systems are required to process this type of information and transform it into meaningful insights. For this study, the theory helps justify why social media analytics tools are needed to interpret

user-generated content, engagement metrics, and digital behaviour patterns.

Without the principles of Big Data Analytics, the scale and complexity of social media data would be unmanageable.

The second theory that underpins this study is Social Network Theory, which examines how individuals interact within networks and how information flows across connected groups. According to He, Zha and Li (2013), social networks contain nodes (users) and ties (relationships), which influence how content spreads and how users respond online. This theory is relevant because analytics tools often map interactions, identify influencers, track communication pathways, and analyse engagement patterns across platforms. Social Network Theory provides a foundation for understanding why some posts go viral, how communities form online, and how users shape brand visibility and perception.

A third theory relevant to the study is the Technology Acceptance Model (TAM). Developed by Davis (1989), TAM explains how users come to accept and use a new technology based on two factors: perceived usefulness and perceived ease of use. This theory is essential for understanding why organizations adopt or reject social media analytics tools. If businesses believe that analytics tools improve decision-making and are easy to use, they will be more likely to adopt them. On the other hand, if tools are seen as complex, expensive, or requiring advanced

technical skills, adoption becomes limited. Since many Ghanaian businesses face technological skill gaps and limited resources, TAM provides a useful explanation for the challenges in adopting analytics tools.

Finally, the Information Systems Success Model by DeLone and McLean (2003) supports the study by explaining what makes an information system effective. The model highlights factors such as system quality, information quality, service quality, system use, and user satisfaction. This theory is applicable because social media analytics tools must produce accurate, reliable, and actionable insights for organizations to find them valuable. If the tool provides misleading results or is difficult to use, it reduces the likelihood of organizations integrating it into their decision-making processes.

Overall, these theories collectively strengthen the foundation of the study. Big Data Analytics Theory explains the nature of the data being analysed, Social Network Theory explains user interactions and engagement patterns, TAM describes the factors influencing tool adoption, and the Information Systems Success Model clarifies what determines successful use of an analytics system. Together, they provide a comprehensive lens for analysing the role, effectiveness, and challenges of social media analytics tools in organizational settings.

2.2.SUMMARY OF THEORETICAL MODELS

Theoretical Framework: This study is guided by several theoretical models that explain how social media data is generated, interpreted, and applied to support decision-making. These models provide a foundation for understanding the function and usefulness of analytics tools for organizations.

Big Data Analytics Theory focuses on analysing large and complex datasets. Social media content fits the “5Vs” of Big Data: volume, velocity, variety, veracity, and value (Rocco et al., 2020). This theory explains how analytics tools process vast amounts of user-generated data in real time to generate meaningful insights.

Social Network Theory examines interactions within networks and their influence on communication patterns. Online networks consist of nodes (users) and connections (interactions) (He, Zha & Li, 2013). Analytics tools often map these interactions, identify influencers, and analyze engagement, helping organizations understand content spread and user responses.

Technology Acceptance Model (TAM) explains how users accept and use new technologies based on perceived usefulness and ease of use (Davis, 1989). This model helps understand why some organizations adopt social media analytics tools while others do not, depending on their perception of benefits and usability.

Information Systems Success Model (DeLone & McLean, 2003) argues that system success depends on system quality, information quality, service quality, use, and user satisfaction. For social media analytics tools, accurate, reliable, and

timely data is essential for organizations to achieve intended benefits.

LITERATURE REVIEW

2.3 Introduction to the Literature Review

This part of the study looks at what different authors and researchers have discussed about social media platforms, the kind of data they generate, and the tools used to analyse that data. The aim is to build a clear understanding of how social media data analytics works and why many organizations now depend on analytics tools to guide decision-making. The review uses information from books, journal articles, and industry reports related to digital marketing, analytics, and information systems. According to Saunders, Lewis and Thornhill (2019), reviewing existing literature is important because it helps place a study within what is already known and highlight gaps that still need attention. By examining earlier studies, this part of the research helps establish the foundation for the current project and shows how it connects to existing knowledge

A review of existing literature reveals significant research on social media analytics and its impact on organizational decision-making.

Stieglitz et al. (2018) investigated how organizations use social media data to support real-time decision-making. Their research focused on understanding the value of social media as a source of organizational intelligence. Using a systematic literature review of over 150 academic studies, they analysed different approaches to social media monitoring and analytics. Their findings showed that although social media provides rich behavioural data, organizations often struggle with the high volume and speed of this data, as well as limited analytical expertise, which affects their ability to make timely decisions.

Boadi, He and Bosompem (2022) focused on the adoption of social media analytics among small and medium-sized enterprises in Ghana. Their methodology involved a mixed-methods approach, consisting of surveys and semi-structured interviews with 120 local businesses. The study found that Ghanaian firms rarely use advanced analytics tools, relying instead on simple metrics like likes and comments. The researchers concluded that high software costs, lack of training, and weak digital strategies prevent organizations from fully leveraging social media data.

Chae (2015) examined how unstructured social media data can be converted into meaningful insights for business decision-making. The research focused on identifying analytical techniques capable of handling messy, diverse digital

content. Using content analysis of real social media datasets, Chae demonstrated that manual interpretation is not practical because data formats vary widely from text to videos, images, and emojis. The study found that analytics tools are essential for performing sentiment analysis, trend detection, and filtering irrelevant information.

Rocco et al. (2020) explored how Big Data Analytics principles apply to social media data interpretation. Their research focus was on understanding how organizations manage the 5Vs of Big Data volume, velocity, variety, veracity, and value. Using conceptual analysis supported by industry case studies, they examined how companies implement Big Data tools. The findings showed that organizations using Big Data systems achieve better audience insights and stronger competitive advantages compared to those relying on traditional methods.

He, Zha and Li (2013) researched how Social Network Theory can be used to analyse online interactions and user influence. Their study focused on mapping social connections and identifying influential users who shape online conversations. Using a network analysis methodology supported by graph-based algorithms, the researchers analysed large datasets from social networks. Their findings revealed that information flow on social media is highly dependent on key influencers and that understanding these relationships can help organizations improve the reach of their content.

Wang and Zhang (2023) conducted a comparative analysis of major social media analytics tools, including Google Analytics, Sprout Social, and Hootsuite. Their research focused on evaluating technical features, usability, and reporting accuracy. Using usability testing and technical assessment as their methodology, they compared dashboard design, response time, integration capabilities, and user experience. Their findings showed that while most tools offer similar analytical features, adoption rates depend heavily on ease of use and system accessibility.

Kaplan and Haenlein (2010) studied the role of social media in digital marketing and customer brand interactions. Their research focused on understanding how social media changes communication dynamics between businesses and audiences. Using theoretical modelling and communication analysis, they examined social media's impact on marketing strategies. They found that timely insights are critical for engagement, and delayed responses weaken customer relationships and brand reputation.

Kietzmann et al. (2011) researched how organizations can strategically manage social media by developing the "honeycomb model." Their study focused on identifying the functional elements that shape user behaviour online. Using conceptual framework development as their methodology, they analyzed seven core building blocks identity, conversations, sharing, presence, relationships, reputation, and groups. Their findings showed that organizations must understand

these building blocks in order to design effective analytics and engagement strategies.

Tuten and Solomon (2017) examined how analytics dashboards support digital marketing decision-making. Their research focused on how marketers use dashboards to interpret performance data. Using case studies of digital marketing campaigns, the researchers analysed how organizations track engagement, identify customer preferences, and optimize campaign execution. Their findings confirmed that dashboards improve decision-making by providing clear, real-time insights that help marketers refine their strategies.

Liu et al. (2016) explored sentiment analysis as a tool for understanding public opinion on social media. Their research focused on how machine learning can classify emotions and attitudes expressed online. Using supervised machine learning techniques applied to large Twitter datasets, the study trained algorithms to detect positive, negative, and neutral sentiments. Their findings showed that sentiment analysis is highly effective for predicting public reactions, monitoring brand perception, and identifying issues early.

2.4.DEFINITIONS OF VARIABLES

In this study, the main variables relate to the concepts, tools, and processes involved in analyzing social media data. Defining these variables helps establish clarity and ensures that the study remains focused on its core objectives.

1. SOCIAL MEDIA

Social media refers to online platforms that allow users to create, share, and interact with content in digital communities. Examples include Facebook, Instagram, TikTok, YouTube, LinkedIn, and X (formerly Twitter). These platforms generate large volumes of user-generated content such as comments, likes, shares, and multimedia posts.

2. SOCIAL MEDIA DATA

This refers to the information generated by user interactions on social media platforms. It includes engagement metrics (likes, comments, shares), follower behaviour, user sentiments, hashtags, keywords, and post performance. Social media data is typically unstructured and produced in real time.

3. SOCIAL MEDIA DATA ANALYTICS

Social media data analytics is the process of collecting, analyzing, and interpreting

data from social media platforms to gain insights into user behaviour, campaign performance, sentiment, and trends. It involves techniques such as content analysis, sentiment analysis, trend detection, and engagement measurement.

4. ANALYTICS TOOLS

Analytics tools are software applications used to process and interpret social media data. Examples include Google Analytics, Hootsuite, Sprout Social, Brandwatch, and Meta Business Suite. These tools provide dashboards, visualizations, and performance metrics that help organizations understand their online audience and make data-driven decisions.

5. ENGAGEMENT METRICS

Engagement metrics are variables used to measure user interaction with content. They include likes, comments, shares, reactions, mentions, and clicks. These metrics help determine how users respond to content and how widely it spreads.

6. SENTIMENT ANALYSIS

Sentiment analysis is the process of determining whether user-generated content such as comments or posts expresses positive, negative, or neutral emotions. It helps organizations understand public opinion and customer attitudes.

7. DASHBOARD VISUALIZATION

Dashboard visualization refers to the graphical presentation of analysed data using charts, graphs, and summaries. Analytics dashboards help users quickly interpret performance indicators and engagement trends.

8. ORGANIZATIONAL DECISION MAKING

This variable refers to the process by which businesses evaluate information and choose strategies. In this study, decision-making relates to how insights from analytics tools influence marketing strategies, content planning, customer engagement, and brand management.

9. CHALLENGES IN ANALYTICS ADOPTION

This variable refers to obstacles businesses face when using social media analytics tools, such as lack of skills, high subscription costs, data privacy concerns, and limited technical knowledge

OVERALL RELEVANCE TO THE STUDY

These theories collectively explain how social media data is produced, how it flows through networks, how it can be interpreted using analytics tools, and what determines whether organizations adopt and benefit from such tools. They form the conceptual backbone of the study by showing why analytics is necessary, how it works, and what factors influence successful implementation.

2.5.SOCIAL MEDIA PLATFORMS AND DATA GENERATION

Social Media Data and Analytics Social media platforms such as Facebook, Instagram, TikTok, YouTube, Linked In, and X (formerly Twitter) are major environments where users share content and interact daily. Each comment, post, like, share, and hashtag contributes to the vast amount of data generated online every moment. According to Stieglitz et al. (2018), social media is a significant source of Big Data due to the speed and volume of content creation, providing insights into trends, opinions, and user behaviour.

Most social media data is unstructured, including text, images, videos, emojis, and reactions, making manual analysis difficult (Chae, 2015). Organizations therefore rely on social media analytics tools to process, organize, and interpret these large datasets.

Engagement metrics like likes, comments, shares, and hashtags help businesses understand audience responses, customer satisfaction, and trending topics (Zhang & Vos, 2014). Timing of user activity also informs when content should be posted for maximum engagement.

Given the rapid and continuous nature of social media activity, analytics tools are essential for tracking behaviour, measuring campaign success, and responding to brand-related issues. As Stieglitz, Mirbabaie, and Böhm (2019) suggest, effective analysis of social media data enhances communication strategies and supports better business

decisions, making these tools critical for organizations seeking to leverage digital insights.

2.6.SOCIAL MEDIA DATA ANALYTICS TOOLS

Social media data analytics tools are software systems designed to collect, process, and interpret data generated from online platforms. These tools help organizations understand how people engage with their posts, what topics are trending, and how their online presence is performing. According to Kapoor et al. (2018), these tools play an important role in modern communication because they help businesses make sense of the huge amount of user activity happening on social media every second.

Examples of commonly used analytics tools include Google Analytics, Hootsuite, Sprout Social, Brandwatch, and Meta Business Suite. Each of these tools offers different features, but they all aim to help users track engagement, measure performance, and analyse user behaviour. For instance, Google Analytics allows organizations to see how users interact with their website after coming from social media, while tools like Hootsuite and Sprout Social help users schedule posts and monitor engagement across multiple platforms. Brandwatch focuses more on social listening by analysing trends, keywords, and conversations happening online (He, Zha & Li, 2013).

One major strength of analytics tools is their ability to organise large amounts of unstructured data and present it in a simple and visual format. As highlighted by Ratten (2020), dashboards, charts, and graphs make it easier for businesses to interpret complex information and make decisions quickly. These visualisations help organizations identify which content performs well and what type of posts gain the most attention from users.

Some tools also offer features for sentiment analysis, which helps determine whether online comments or discussions are positive, negative, or neutral. This is useful for understanding customer attitudes toward a brand or product. Other tools track keywords and hashtags to find trending topics, allowing organizations to join relevant conversations and increase visibility.

Even though the tools have different strengths, they all share a common purpose: to transform raw social media data into insights that support decision-making. Without them, it would be extremely difficult for businesses to keep up with the speed and volume of digital information being generated online.

2.7.BENEFITS OF SOCIAL MEDIA ANALYTICS

Social media analytics helps organizations understand their audience by analyzing likes, comments, shares, and viewing patterns, providing insights into customer preferences and behavior (Arora et al., 2019). It also measures campaign performance, giving precise data on engagement, reach, and visibility, which allows businesses to adjust strategies quickly (Tuten & Solomon, 2020).

Analytics supports data-driven decision-making by highlighting audience concerns, trends, and online conversations (Guesalaga, 2016). It improves customer engagement, enabling timely responses to messages and recurring issues, which strengthens relationships (Kapoor et al., 2018). Additionally, analytics identifies opportunities by tracking trending topics and competitor performance, helping organizations capitalize on market gaps. Overall, social media analytics equips businesses with essential insights to improve performance and make informed decisions.

2.8.CHALLENGES OF USING SOCIAL MEDIA ANALYTICS

Even though social media analytics tools provide many benefits, organizations still face several challenges when trying to use them effectively. One of the most common challenges is the lack of technical skills. Many businesses, especially small and medium-sized ones, do not have staff who understand how to interpret analytics reports or work with complex dashboards. According to Chaffey and Ellis-Chadwick (2019), many organizations struggle to take full advantage of analytics tools because their employees lack the necessary digital skills.

Another challenge is the cost of advanced analytics tools. While some platforms offer free versions, the more detailed features often require paid subscriptions, which can be expensive for smaller organizations. Guesalaga (2016) explains that the financial cost of analytics tools can limit their adoption, especially in developing markets where marketing budgets are already low.

Data privacy and security concerns also create challenges. Organizations must be careful when collecting or analysing user information, especially with global privacy laws such as GDPR. Stieglitz et al. (2018) note that concerns about user

privacy and platform restrictions can affect how much data an organization is able to access or analyse.

In addition, the overwhelming volume of social media data can make it difficult for users to determine what information is relevant. Kapoor et al. (2018) mention that organizations often face problems filtering out unnecessary data, which makes it harder to extract meaningful insights.

Another challenge is the fast-changing nature of social media platforms. Features, algorithms, and trending topics change constantly, making it difficult for businesses to keep up. As Tuten and Solomon (2020) point out, platforms frequently update their systems, which means analytics tools must also change to stay accurate. This constant change can cause confusion or outdated reports if tools are not updated regularly.

Finally, some organizations struggle with integrating analytics insights into their actual decision-making processes. Even when the tools provide useful information, companies may not have clear strategies for how to use the insights. This leads to wasted data and missed opportunities.

Overall, while social media analytics tools have great potential, challenges such as skill gaps, cost, privacy issues, overwhelming data, platform changes, and weak

integration into decision-making processes can prevent organizations from using them effectively.

2.9. GAPS IN LITERATURE

Research Gaps

Despite extensive studies on social media platforms and analytics tools, several gaps remain. Most research focuses on large international companies, neglecting smaller businesses, particularly in developing countries, which face unique challenges like limited budgets and digital skills (Guesalaga, 2016). Additionally, while many studies discuss tool features and benefits, fewer examine how organizations apply analytics insights in practice (Kapoor et al., 2018).

There is limited research on local contexts, such as Ghana, where technology adoption is influenced by cost, training, and infrastructure (Boadi, He & Bosompem, 2022). Technical limitations of analytics tools, including data inaccuracies and restricted access, are also underexplored (Stieglitz et al., 2018). Finally, few studies address how academic or prototype systems compare to commercial tools, leaving a gap in understanding educational or simplified analytics systems. These gaps highlight the need for research on small organizations, local contexts, practical application, and tool limitations.

CHAPTER THREE

METHODOLOGY

3.0.Introduction

This chapter explains the approach used to develop the social media data analytics prototype system. It describes the software development methodology chosen, the reason for selecting it, and the steps followed throughout the development process. The chapter also outlines the tools, technologies, system requirements, design activities, implementation steps, and testing procedures used to ensure the system works as intended. The purpose of this chapter is to give a clear understanding of how the system was planned, built, and evaluated.

3.1. SOFTWARE DEVELOPMENT METHODOLOGY

For this project, the Waterfall Model was used as the software development approach. The Waterfall Model is a structured method where each phase is completed before moving to the next one. Sommerville (2016) notes that the Waterfall approach works well when system requirements are clear and stable from the beginning. Since the prototype system has well-defined features such as data upload, engagement calculation, and dashboard visualization the Waterfall method is suitable and easy to follow.

The Waterfall Model also allows the researcher to work in an organised manner by completing each stage step-by-step. Pressman and Maxim (2015) explain that this method helps avoid confusion and ensures that every phase is properly documented. This makes it ideal for an academic project like this, where the aim is to demonstrate understanding of both system design and development.

3.2 PHASES OF THE METHODOLOGY

Requirements Gathering

At this stage, the functional and non-functional requirements of the system is identified. Functional requirements included features such as uploading social media datasets, processing engagement metrics, and displaying results on a dashboard. Non-functional requirements focused on simplicity, ease of use, and quick data processing. As Pressman and Maxim (2015) explain, clearly defining requirements is important because it guides the entire development process and ensures the system fulfils its purpose.

System Design

The design stage involves creating diagrams and sketches that show how the system will work. This includes designing the user interface, system architecture, and data flow. Sommerville (2016) states that system design provides a blueprint that developers can follow during implementation. For this project, wireframes were created to show the dashboard layout, and flowcharts were used to map how uploaded data moves through the system.

Implementation

During implementation, the system will be developed using appropriate programming languages and tools. HTML, CSS, and JavaScript were used for the user interface, while Python was used for processing the uploaded social media data. Pressman and Maxim (2015) note that implementation involves turning design documents into an actual working system. In this stage, the system's main modules data upload, processing script, and dashboard—will be coded and integrated.

To keep the system simple and aligned with project goals, it relied on **manually uploaded datasets** rather than real-time social media API connections. This ensured that the system remained easy to test and demonstrate within the available time.

Testing

After implementation, the system is tested to confirm that all functions work correctly. Testing include checking if data could be uploaded successfully, verifying whether engagement metrics are processed accurately, and ensuring that the dashboard displays results properly. Sommerville (2016) emphasises that testing is important because it reveals errors and ensures that the system meets its

requirements. Any issues discovered during testing were fixed before the final demonstration.

Deployment and Demonstration

Once testing is completed, the prototype will be prepared for presentation. This step involved running the system in a controlled environment to show how it processes sample social media data and displays insights. Although the system is not meant for commercial use, this demonstration highlights how organizations can benefit from analyzing online engagement patterns

Documentation

The final stage involved preparing documentation such as system descriptions, design diagrams, and instructions on how to run the prototype. Proper documentation makes the system easy to understand, especially for supervisors and examiners. According to Sommerville (2016), documentation is essential in software development because it ensures the system can be reviewed, improved, or maintained in the future.

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THANK YOU!!!